

SETS — Short Revision Notes

Apply Level: Use Knowledge in New Situations

1. Writing Sets in Different Forms

Method	How To	Example
Descriptive	Describe in words	{Even numbers between 0 and 10}
Roster (Listing)	List elements in { }	{2, 4, 6, 8}
Set-Builder	Use a rule	{ $x \mid x$ is even, $0 < x < 10$ }

Quick Application

Given Information	Write in Roster Form
Multiples of 5 between 0 and 20	{5, 10, 15, 20}
Letters in "RECONCILIATION"	{R, E, C, O, N, I, L, A, T}
Prime numbers between 2 and 13	{2, 3, 5, 7, 11, 13}
Whole numbers between 5 and 10	{6, 7, 8, 9}

2. Finding the Number of Elements $n(A)$

Set	$n(A)$	Why?
{2, 4, 6, 8}	4	4 elements
{5, 10, 15, 20, 25}	5	5 elements
{ }	0	Empty set
{1, 2, 3, 4, 5, 6, 7, 8}	8	8 elements

Quick Application

Set	$n(A)$
{Multiples of 3 between 0 and 20}	6
{Prime numbers less than 10}	4
{Vowels in English alphabet}	5
{Letters in "MATHEMATICS"}	8

3. Identifying Finite and Infinite Sets

Rule: if you can count all elements $\rightarrow$ FINITE; if counting never ends $\rightarrow$ INFINITE.

Set	Type	Why?
{Multiples of 5 from 0 to 50}	FINITE	{5,10,...,45,50} — countable
{Integers}	INFINITE	..., -2, -1, 0, 1, 2, ...
{Digits of 25265}	FINITE	{2,5,6} (3 elements)
{Positive integers that are not prime}	INFINITE	1,4,6,8,9,10,12,...

4. Finding Equal and Equivalent Sets

Equal Sets — exactly the SAME elements

Set A	Set B	Equal?
{2,4,6,8}	{4,8,2,6}	YES ✓ (same elements)
{3,6,9}	{6,9,3}	YES ✓
{1,2,3}	{1,2,3,4}	NO ✗

Equivalent Sets — the SAME number of elements

Set A	Set B	Equivalent?	Why?
{2,4,6,8}	{a,e,i,o}	YES ✓	Both have 4 elements
{3,6,9}	{1,2,3,4,5}	NO ✗	$3 \neq 5$

$P = \{3, 6, 9\}$

$Q = \{\text{letters of "net"}\} = \{n, e, t\} = \{e, n, t\}$

$R = \{\text{odd numbers between 0 and 10}\} = \{1, 3, 5, 7, 9\}$

Equal: $Q = \{\text{letters of "ten"}\}$ (same elements $\{e, n, t\}$)

Equivalent: P and $Q \rightarrow$ YES (both have 3 elements)

P and $R \rightarrow$ NO (3 vs 5 elements)

5. Writing Subsets

Formula: number of subsets = 2^n , where $n = n(A)$.

Set	n	Subsets	All Subsets
{ }	0	1	{ }
{a}	1	2	{ }, {a}
{a,b}	2	4	{ }, {a}, {b}, {a,b}
{a,b,c}	3	8	{ }, {a}, {b}, {c}, {a,b}, {a,c}, {b,c}, {a,b,c}

Quick Application

Set	Subsets	Write Some Subsets
{2,4,6}	8	{ }, {2}, {4}, {6}, {2,4}, {2,6}, {4,6}, {2,4,6}
{red, blue}	4	{ }, {red}, {blue}, {red, blue}
{1,2,3,4}	16	{ }, {1}, {2}, {3}, {4}, {1,2}, ...

6. Applying Intersection ($A \cap B$)

Rule: elements COMMON to both sets.

Set A	Set B	$A \cap B$
{2,4,6,8}	{1,2,3,4}	{2,4}
{1,3,5,7}	{2,4,6,8}	{ } (disjoint)
{a,b,c,d}	{c,d,e,f}	{c,d}
{1,2,3,4,5}	{4,5,6,7,8}	{4,5}

$P = \{1, 3, 6, 8, 10, 13\}$ $Q = \{1, 6, 7, 8\}$ $R = \{2, 3, 9, 10, 12\}$

$$P \cap Q = \{1, 6, 8\}$$

$$P \cap R = \{3, 10\}$$

$$Q \cap R = \{ \}$$

7. Applying Union ($A \cup B$)

Rule: all elements in EITHER or BOTH sets (no duplicates).

Set A	Set B	$A \cup B$
{2,4,6,8}	{1,2,3,4}	{1,2,3,4,6,8}
{1,3,5,7}	{2,4,6,8}	{1,2,3,4,5,6,7,8}
{a,b,c}	{c,d,e}	{a,b,c,d,e}

$$P = \{1, 3, 6, 8, 10, 13\} \quad Q = \{1, 6, 7, 8\} \quad R = \{2, 3, 9, 10, 12\}$$

$$P \cup Q = \{1, 3, 6, 7, 8, 10, 13\}$$

$$P \cup R = \{1, 2, 3, 6, 8, 9, 10, 12, 13\}$$

$$Q \cup R = \{1, 2, 3, 6, 7, 8, 9, 10, 12\}$$

8. Applying Complement (A')

Rule: elements in ϵ NOT in A.

Universal Set ϵ	Set A	A'
{1,2,3,4,5,6,7}	{2,4,6}	{1,3,5,7}
{5,10,15,20,25,30,35}	{10,20,30}	{5,15,25,35}
{a,b,c,d,e,f}	{a,c,e}	{b,d,f}

$$\epsilon = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$P = \{1, 3, 5, 7, 9, 10\} \quad \rightarrow \quad P' = \{2, 4, 6, 8\}$$

$$Q = \{2, 4, 5, 7, 8\} \quad \rightarrow \quad Q' = \{1, 3, 6, 9, 10\}$$

9. Applying Set Relationships in Venn Diagrams

Example — reading a Venn diagram

Region	Elements	Set
In A only	{2,6}	A only
In B only	{1,3,7}	B only
In both ($A \cap B$)	{4}	$A \cap B$
Outside both	{5,8,9,10}	$(A \cup B)'$

Finding $n(A)$ from the Venn diagram

Set	Elements	$n(A)$
A	{2,4,6}	3
B	{1,3,4,7}	4
$A \cap B$	{4}	1
$A \cup B$	{1,2,3,4,6,7}	6
A'	{1,3,5,7,8,9,10}	7

10. Applying Complex Set Problems

Example 1 — find elements

Given $\varepsilon = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $P = \{2, 4, 6\}$, $Q = \{1, 5, 8\}$:

Find	Solution
$P \cap Q$	$\{ \}$ (no common elements)
$P \cup Q$	$\{1, 2, 4, 5, 6, 8\}$
P'	$\{1, 3, 5, 7, 8\}$
Q'	$\{2, 3, 4, 6, 7\}$

Example 2 — given the Venn diagram, find the sets

Region	Elements	Set
Only X	$\{2, 8, 10\}$	X only
Only Y	$\{1, 6, 7\}$	Y only
Both X and Y	$\{4, 9\}$	$X \cap Y$
Neither	$\{3, 5\}$	$(X \cup Y)'$

11. Application Checklist

Step 1 - Understand what is given:

- ☐ What is the universal set (ε)?
- ☐ What are the sets given?
- ☐ Descriptive, roster, or Venn diagram form?

Step 2 - Identify what to find:

- ☐ $A \cap B$? $A \cup B$? A' ? $n(A)$?
- ☐ Equal / equivalent? Subset?

Step 3 - Apply the correct operation:

- ☐ Intersection ($\cap$) -> common elements
- ☐ Union ($\cup$) -> all elements (no duplicates)
- ☐ Complement ($'$) -> elements in ε not in A
- ☐ Subset -> check all elements are in the other set

Step 4 - Write the answer:

- ☐ In roster form: {elements}
- ☐ With correct notation

12. Self-Test — Application Questions

No.	Question	Answer
1	Write {multiples of 5 between 0 and 20} in roster form	{5,10,15,20}
2	Find $n(\{2,4,6,8,10\})$	5
3	Is {3,6,9} finite or infinite?	Finite
4	Are {1,2,3} and {3,2,1} equal?	Yes
5	Are {1,2,3} and {4,5,6} equivalent?	Yes (both have 3 elements)
6	How many subsets does {a,b,c} have?	8
7	$A=\{2,4,6,8\}$, $B=\{1,2,3,4\}$. Find $A \cap B$	{2,4}
8	$A=\{2,4,6,8\}$, $B=\{1,2,3,4\}$. Find $A \cup B$	{1,2,3,4,6,8}
9	$\epsilon=\{1,2,3,4,5,6,7\}$, $A=\{2,4,6\}$. Find A'	{1,3,5,7}
10	$A=\{1,3,5\}$, $B=\{2,4,6\}$. Are they disjoint?	Yes

Revision Tip: Practice applying set operations to different examples. Start with simple sets, then move to Venn diagrams — and always check your answers by verifying the elements.